

# What is an Experiment?

## Definition (Random Experiment)

A **random experiment** (or statistical experiment) is a process, trial, or observation that leads to one of several possible outcomes, where the exact result cannot be predicted with absolute certainty beforehand.

## Three Key Characteristics of a Random Experiment

- 1 **Repeatability:** The process can, at least in theory, be repeated indefinitely under identical conditions.
- 2 **Unpredictability:** While the conditions are controlled, the outcome of any single trial is governed by chance.
- 3 **Predefined Outcome Space:** Although we do not know *which* specific outcome will occur, the set of all possible outcomes is fully known before the trial begins.

# Defining the Outcome Space: The Sample Space

## Definition (Sample Space)

The **sample space** of a random experiment, denoted by  $\mathcal{S}$  (or the Greek letter  $\Omega$ ), is the set of all possible outcomes of that experiment. An individual result in this set is called an **outcome** or a **sample point**.

## Classification of Sample Spaces

Depending on the nature of the experimental outcomes,  $\mathcal{S}$  can be classified as:

- **Discrete (Finite or Countable):** Outcomes can be listed in a sequence.
- **Continuous (Uncountable):** Outcomes form an interval of real numbers, representing measurements.

# Defining Events: Subsets of the Sample Space

## Definition (Event)

An **event** is any collection of outcomes of a random experiment.

Mathematically, an event  $E$  is a **subset** of the sample space  $\mathcal{S}$  (denoted by  $E \subseteq \mathcal{S}$ ).

An event  $E$  is said to **occur** if the actual observed outcome of the experiment is a member of the set  $E$ .

## Special Cases of Events

- **Simple (Elementary) Event:** An event containing exactly one outcome (e.g.,  $E = \{e\}$  where  $e \in \mathcal{S}$ ).
- **Sure (Certain) Event:** The entire sample space  $\mathcal{S}$  itself. It is guaranteed to occur on every trial.
- **Null (Impossible) Event:** The empty set  $\emptyset$ . It contains no outcomes and can never occur.

# Examples

## Examples

- **Rolling a Fair Six-Sided Die:**  $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ 
  - Event  $A$  (Rolling an even number):  $A = \{2, 4, 6\}$
  - Event  $B$  (Rolling a number greater than 4):  $B = \{5, 6\}$
- **Tossing a Coin Twice:**  $\mathcal{S} = \{HH, HT, TH, TT\}$ 
  - Event  $C$  (Getting exactly one head):  $C = \{HT, TH\}$
  - Event  $D$  (Getting the same result on both tosses):  $D = \{HH, TT\}$
- **Lifetime of a Chip ( $X$  hours):**  $\mathcal{S} = \{x \in \mathbb{R} \mid x \geq 0\}$ 
  - Event  $E$  (The chip lasts more than 1000 hours):  
 $E = \{x \in \mathcal{S} \mid x > 1000\}$
  - Event  $F$  (The chip fails within the first 100 hours):  
 $F = \{x \in \mathcal{S} \mid 0 \leq x \leq 100\}$

# Set Operations on Events

## Definition (Basic Operations)

Let  $A$  and  $B$  be events in a sample space  $\mathcal{S}$ . We define:

- **Union** ( $A \cup B$ ): The event that occurs if  $A$  *or*  $B$  (or both) occur.
- **Intersection** ( $A \cap B$ ): The event that occurs if both  $A$  *and*  $B$  occur.
- **Complement** ( $A^c$  or  $A'$ ): The event that occurs if  $A$  does *not* occur.

## Mutually Exclusive (Disjoint) Events

Two events  $A$  and  $B$  are said to be **mutually exclusive** or **disjoint** if they cannot occur at the same time. Mathematically, this is expressed as:

$$A \cap B = \emptyset$$

# Examples

## Examples

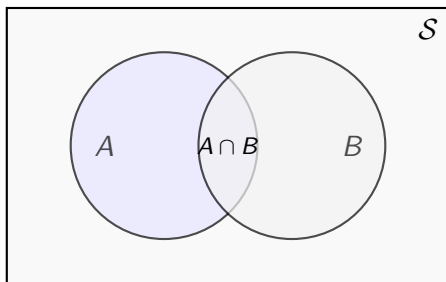
- **Rolling a Die:** Let  $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ . If  $A = \{2, 4, 6\}$  (even) and  $B = \{1, 3, 5\}$  (odd):
  - $A \cap B = \emptyset$ , meaning the events are mutually exclusive.
- **System Status Monitor:** Suppose a system is monitored and its state space is  $\mathcal{S} = \{\text{Active}, \text{Idle}, \text{Overloaded}\}$ . If  $E_1 = \{\text{Active}\}$  and  $E_2 = \{\text{Active}, \text{Overloaded}\}$ :
  - $E_1 \cup E_2 = \{\text{Active}, \text{Overloaded}\}$
  - $E_1 \cap E_2 = \{\text{Active}\}$

# Visualizing Events: Venn Diagrams

## Venn Diagrams

Venn Diagrams visually represent the relationship between events in a sample space  $\mathcal{S}$ :

- **Sample Space ( $\mathcal{S}$ ):** Represented by the outer bounding rectangle.
- **Events ( $A, B$ ):** Represented by closed regions (usually circles or ellipses).



# De Morgan's Law

These laws establish fundamental mathematical dualities between unions, intersections, and complements.

## 1. Complement of a Union:

$$(A \cup B)^c = A^c \cap B^c$$

- *Intuition:* The event that "neither  $A$  nor  $B$  occurs" is identical to " $A$  does not occur" **and** " $B$  does not occur".

## 2. Complement of an Intersection:

$$(A \cap B)^c = A^c \cup B^c$$

- *Intuition:* The event that " $A$  and  $B$  do not both occur" is equivalent to " $A$  does not occur" **or** " $B$  does not occur".

# Exercise 1: Cardinality & Two Fair Dice

## Problem Statement

Two fair, six-sided dice are rolled. The sample space contains  $|\mathcal{S}| = 36$  equally likely outcomes. Let:

- Event  $A$ : The sum of the two dice is exactly 7.
- Event  $B$ : At least one of the dice shows a 4.

Find:

(a)  $|A|$ , (b)  $|B|$ , (c)  $|A \cap B|$ , and (d)  $|A \cup B|$ .

## Exercise 2: Card Deck & De Morgan's Laws

### Problem Statement

A single card is drawn from a standard, well-shuffled 52-card deck ( $|\mathcal{S}| = 52$  equally likely outcomes). Let:

- Event  $F$ : The drawn card is a Face Card (Jack, Queen, or King).
- Event  $H$ : The drawn card is from Heart suit.

Find: (a)  $|F \cup H|$ , and (b) the number of ways that the card is **neither** a face card **nor** a heart using De Morgan's Laws and complement cardinality.

## Exercise 3: Three Coin Tosses

### Problem Statement

A fair coin is tossed three times.

- 1 Write out the complete, equally likely sample space  $S$ .
- 2 Let Event  $A$  be "obtaining at least two heads" and Event  $B$  be "obtaining exactly one tail". Find  $|A|$ ,  $|B|$ , and  $|A \cap B|$ .

# What is Probability?

## Definition (Probability)

Historically and mathematically, **probability** is a numerical measure (ranging from 0 to 1) assigned to an event that quantifies the likelihood of its occurrence.

- A probability of 0 indicates the event is **impossible**.
- A probability of 1 indicates the event is **certain**.

## Paradigm 1: The Classical Definition (Equally Likely)

If a random experiment has a finite sample space  $\mathcal{S}$  consisting of  $N$  outcomes, and we assume all outcomes are **equally likely** to occur:

$$\text{Probability of Event } A = P(A) = \frac{\text{Number of outcomes in } A}{\text{Total number of outcomes in } \mathcal{S}} = \frac{|A|}{|\mathcal{S}|}$$

# Quantifying Probability: Empirical & Subjective

## Paradigm 2: The Empirical Definition (Relative Frequency)

When outcomes are not equally likely, we estimate probabilities by conducting a random experiment repeatedly. If the experiment is performed  $n$  times, and event  $B$  occurs  $n_B$  times:

$$P(B) \approx \frac{n_B}{n}$$

As the number of trials grows infinitely large ( $n \rightarrow \infty$ ), this relative frequency stabilizes to the true probability (formalized later by the *Law of Large Numbers*).

# Examples

## Classical Example

Consider rolling a fair, six-sided die. The sample space is  $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ , so  $|\mathcal{S}| = 6$ . Let  $A$  be the event of rolling an even number,  $A = \{2, 4, 6\}$ , so  $|A| = 3$ .

$$P(A) = \frac{3}{6} = 0.5$$

## Empirical Example

In testing a manufacturing assembly line, we observe that out of 1,000 manufactured items, 15 are defective.

$$\text{Estimated Probability of a Defect} = \frac{15}{1000} = 0.015 \text{ (or 1.5\%)}$$

# Comparing Approaches: Equally Likely vs. General

## Comparison between two definitions

The primary difference between these paradigms is the assumption of symmetry.

| Feature              | Classical                    | Empirical/General      |
|----------------------|------------------------------|------------------------|
| <b>Assumption</b>    | Uniform (All outcomes equal) | Non-uniform (Weighted) |
| <b>Condition</b>     | Symmetry exists              | No symmetry required   |
| <b>Calculation</b>   | Simple Ratio: $ A / S $      | Frequency: $n_A/n$     |
| <b>Applicability</b> | Idealized experiments        | Real-world data        |

Despite these differences both of these approaches share some fundamental properties.

## Property 1: Boundedness ( $0 \leq P(A) \leq 1$ )

### The Property

The probability of any event  $A$  is restricted to a value between 0 and 1.

### Classical Derivation (Set Cardinality)

Since  $A \subseteq \mathcal{S}$ , the size of  $A$  cannot be negative or exceed the total size of  $\mathcal{S}$ :

$$0 \leq |A| \leq |\mathcal{S}| \implies 0 \leq \frac{|A|}{|\mathcal{S}|} \leq 1 \implies 0 \leq P(A) \leq 1$$

### Empirical Derivation (Trial Counting)

The count of occurrences  $n_A$  in  $n$  trials must lie between 0 and  $n$ :

$$0 \leq n_A \leq n \implies 0 \leq \frac{n_A}{n} \leq 1 \implies 0 \leq f_n(A) \leq 1$$

## Property 2: Certainty & Impossibility

### The Property

The certain event (full sample space) has probability 1; the impossible event (empty set) has probability 0.

$$P(\mathcal{S}) = 1 \quad \text{and} \quad P(\emptyset) = 0$$

### Derivation

For the sure event  $A = \mathcal{S}$  and the impossible event  $A = \emptyset$ :

$$P(\mathcal{S}) = \frac{|\mathcal{S}|}{|\mathcal{S}|} = 1 \quad \text{and} \quad P(\emptyset) = \frac{|\emptyset|}{|\mathcal{S}|} = \frac{0}{|\mathcal{S}|} = 0$$

$$f_n(\mathcal{S}) = \frac{n}{n} = 1 \quad \text{and} \quad f_n(\emptyset) = \frac{0}{n} = 0$$

## Property 3: Additivity of Disjoint Events

### The Property

If events  $A$  and  $B$  are **mutually exclusive** ( $A \cap B = \emptyset$ ), then the probability of their union is the sum of their individual probabilities.

$$P(A \cup B) = P(A) + P(B)$$

### Derivation

If  $A$  and  $B$  share no common elements, cardinality is additive:

$|A \cup B| = |A| + |B|$ . Thus,

$$\frac{|A \cup B|}{|S|} = \frac{|A| + |B|}{|S|} = \frac{|A|}{|S|} + \frac{|B|}{|S|} \implies P(A \cup B) = P(A) + P(B)$$

$$f_n(A \cup B) = \frac{n_A + n_B}{n} = \frac{n_A}{n} + \frac{n_B}{n} \implies f_n(A \cup B) = f_n(A) + f_n(B)$$

# Exercises

## Complement Event

Prove that the probability of the complement of any event  $A$  is given by:

$$P(A^c) = 1 - P(A)$$

## Monotonicity

Prove that if event  $A$  is a subset of event  $B$  ( $A \subseteq B$ ), then its probability cannot exceed the probability of  $B$ :

$$P(A) \leq P(B)$$

# Exercises

## General Addition rule

Prove that for any two events  $A$  and  $B$  (which may overlap):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

## Hint

First, prove principle of inclusion-exclusion for counting finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

- Explain why we must subtract the size of the intersection  $|A \cap B|$ .

**Question:** How can you extend this?

# Example

## Problem Statement

$A$  throws six dice and wins if he scores at least one ace.  $B$  throws twelve dice and wins if he scores at least two aces. Who has the greater probability to win?

# Example

## Problem Statement

Consider the 24 possible arrangements (permutations) of the symbols 1234 and assume each arrangement has probability  $\frac{1}{24}$ . Let  $A_i$  be the event that the digit  $i$  appears at its natural place (where  $i = 1, 2, 3, 4$ ).

Find:

- 1  $P(A_i)$
- 2  $P\left(\bigcup_{i=1}^4 A_i\right)$
- 3  $P\left(\bigcap_{i=1}^4 A_i\right)$

## Example

### Problem Statement

A coin is tossed until for the first time the same result appears twice in succession. Describe the sample space. Find the probability of the following events:

- 1 Are the outcomes equally likely?
- 2 The experiment ends before the sixth toss.
- 3 An even number of tosses is required.

### Problem Statement

10 pairs of shoes are in a closet. You choose 4 out of them at random. What is the probability that you have at least one matching pair?

# Bridging to Formal Theory: Why Axioms?

These intuitive properties are consistent across both viewpoints:

| Concept     | Classical Ratio                   | Empirical Count       |
|-------------|-----------------------------------|-----------------------|
| Boundedness | $0 \leq  A / \mathcal{S}  \leq 1$ | $0 \leq n_A/n \leq 1$ |
| Certainty   | $ \mathcal{S} / \mathcal{S}  = 1$ | $n/n = 1$             |
| Additivity  | $( A  +  B )/ \mathcal{S} $       | $(n_A + n_B)/n$       |

## The Limitations

- **Classical definition** falls short when outcomes are not equally likely.
- **Empirical definition** relies on an impossible, unverifiable limit ( $n \rightarrow \infty$ ).

# The Experiment

## The Setup

Suppose an experimenter rolls two fair, six-sided dice.

- The fundamental sample space  $\mathcal{S}$  has 36 equally likely outcomes.

## The Hidden Transformation

Before you can see the dice, they are passed through a digital "black box." The box applies an **unknown function**  $Z = f(x, y)$  to the outcomes of the two dice ( $x$  and  $y$ ).

- You, the observer, are only shown the final output  $Z \in \mathcal{T} = \{0, 1, 2, 3, 4, 5\}$ .

## The Core Questions

- 1 Can we assume that the observable outcomes in  $\mathcal{T}$  are equally likely?
- 2 If you only have access to  $\mathcal{T}$ , can you mathematically calculate their probabilities?

# Revealing the Hidden Function

## The True function

Let us reveal the unknown function programmed inside the black box:

$$Z = f(x, y) = |x - y| \quad (\text{The absolute difference between the dice})$$

## Calculating the Real Probabilities

We must map the outputs back to the original, equally likely 36 outcomes in  $\mathcal{S}$ :

- $Z = 0 \implies \{(1, 1), (2, 2), \dots, (6, 6)\} \implies 6 \text{ outcomes} \implies P(Z = 0) = \frac{6}{36}$
- $Z = 1 \implies \{(1, 2), (2, 1), (2, 3), \dots\} \implies 10 \text{ outcomes} \implies P(Z = 1) = \frac{10}{36}$
- $Z = 5 \implies \{(1, 6), (6, 1)\} \implies 2 \text{ outcomes} \implies P(Z = 5) = \frac{2}{36}$

**Conclusion: They are Not Equally Likely!**

# The Problem

If the mapping function remains unknown, classical Calculation is Impossible

Without knowing the mapping function, it is **mathematically impossible** to calculate the true probabilities of  $\mathcal{T}$  purely from its set definition.

## The Alternative: Empirical Estimation

The only way to estimate the probabilities is to rely on the **empirical (relative frequency) approach**:

- Collect and record an observational stream of outcomes over  $n$  runs.
- Estimate the probability of each outcome as  $f_n(Z) = \frac{n_Z}{n}$ .

## The Catch: An Unknown Limit

Since we can never perform infinite trials, we cannot verify if this limit truly stabilizes, leaving the mathematical reasoning incomplete!

# The Problem

## The Reality

Almost all real-world phenomena we try to model are exactly like this experiment. The complex systems we observe are rarely isolated, simple events.

## Hidden Factors and Confounding Variables

What we measure is the cumulative output of countless underlying physical forces, behaviors, and environmental factors:

- Many of these variables are completely hidden or unknown to us.
- They interact in non-linear ways that we may not be able to model.

## The Motivation for a New Approach

This complexity is precisely why we need a robust, unified definition of probability. To work around the physical limitations of classical and empirical approaches, we need a formal mathematical system that operates independently of our ability to observe every underlying factor.

# Kolmogorov's Three Axioms

Let  $\mathcal{S}$  be the sample space of a random experiment. To every event  $A \subseteq \mathcal{S}$ , we assign a real number  $P(A)$  satisfying the following:

- **Axiom 1: Non-Negativity** For any event  $A$ , the assigned probability cannot be negative:

$$P(A) \geq 0$$

- **Axiom 2: Certainty** The probability of the entire sample space (the sure event) is exactly 1:

$$P(\mathcal{S}) = 1$$

- **Axiom 3: Countable Additivity** For any infinite sequence of mutually exclusive (disjoint) events  $A_1, A_2, A_3, \dots$  (where  $A_i \cap A_j = \emptyset$  for all  $i \neq j$ ):

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

## Consequence of Axioms

### Theorem: The Complement Rule

For any event  $A$ , the probability of its complement is:

$$P(A^c) = 1 - P(A)$$

### Theorem: Probability of the Null Set

The probability of the empty set (impossible event) is zero:

$$P(\emptyset) = 0$$

### Theorem: Monotonicity

If event  $A$  is a subset of event  $B$  ( $A \subseteq B$ ), then:

$$P(A) \leq P(B)$$

## Proof: The Complement Rule

By set definition, an event  $A$  and its complement  $A^c$  are mutually exclusive ( $A \cap A^c = \emptyset$ ) and their union partitions the entire sample space ( $A \cup A^c = \mathcal{S}$ ).

1. Apply **Axiom 3** (for finite disjoint events, where  $A_3, A_4, \dots = \emptyset$ ):

$$P(A \cup A^c) = P(A) + P(A^c)$$

2. Since  $A \cup A^c = \mathcal{S}$ , substitute on the left side:

$$P(\mathcal{S}) = P(A) + P(A^c)$$

3. Apply **Axiom 2** ( $P(\mathcal{S}) = 1$ ):

$$1 = P(A) + P(A^c) \implies P(A^c) = 1 - P(A) \quad \blacksquare$$

## Proof: Probability of the Null Set

Let the event  $A = \mathcal{S}$ . Its complement is the empty set:

$$A^c = \mathcal{S}^c = \emptyset$$

1. Apply the **Complement Rule** ( $P(A^c) = 1 - P(A)$ ) to this setup:

$$P(\emptyset) = P(\mathcal{S}^c) = 1 - P(\mathcal{S})$$

2. Now, substitute **Axiom 2** ( $P(\mathcal{S}) = 1$ ):

$$P(\emptyset) = 1 - 1 = 0 \quad \blacksquare$$

## Proof: Monotonicity

If  $A \subseteq B$ , we can decompose event  $B$  into two disjoint parts: the part inside  $A$ , and the part that is in  $B$  but outside  $A$  (denoted  $B \cap A^c$ ):

$$B = A \cup (B \cap A^c) \quad \text{where} \quad A \cap (B \cap A^c) = \emptyset$$

1. Apply **Axiom 3** (Additivity):

$$P(B) = P(A) + P(B \cap A^c)$$

2. Apply **Axiom 1** (Non-negativity), we know  $P(B \cap A^c) \geq 0$ .

3. Therefore, adding a non-negative term to  $P(A)$  yields:

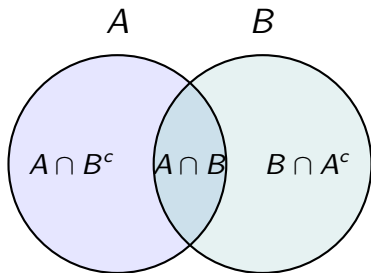
$$P(B) \geq P(A) \quad \blacksquare$$

# The General Addition Rule (Overlapping Events)

## Theorem: General Addition Rule

For any two events  $A$  and  $B$  (not necessarily disjoint):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



## Proof: General Addition Rule

We can partition the union  $A \cup B$  and the event  $B$  itself into mutually exclusive parts:

1. Partition  $A \cup B$ :  $A \cup B = A \cup (B \cap A^c)$ . By Axiom 3:

$$P(A \cup B) = P(A) + P(B \cap A^c) \quad \text{— (Equation 1)}$$

2. Partition  $B$ :  $B = (A \cap B) \cup (B \cap A^c)$ . By Axiom 3:

$$P(B) = P(A \cap B) + P(B \cap A^c)$$

3. Solve the second equation for  $P(B \cap A^c)$ :

$$P(B \cap A^c) = P(B) - P(A \cap B) \quad \text{— (Equation 2)}$$

4. Substitute Equation 2 into Equation 1:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \blacksquare$$

# The Intuition of Conditioning

In many scenarios, we acquire partial information about the outcome of an experiment before its full result is revealed.

## Restricting the Sample Space

When we learn that an event  $B$  has occurred, any outcome outside of  $B$  becomes impossible.

- The original sample space  $\mathcal{S}$  collapses.
- The event  $B$  becomes our **new, effective sample space**.

We must now recalculate the probability of an event  $A$  given this restricted domain. This is denoted as  $P(A \mid B)$ .

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# Formal Definition of Conditional Probability

## Definition

Conditional Probability Let  $A$  and  $B$  be two events in a sample space  $\mathcal{S}$  where  $P(B) > 0$ . The conditional probability of  $A$  given that  $B$  has occurred is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

## Immediate Operational Takeaway:

- The formula is mathematically undefined if  $P(B) = 0$ . We can only condition on events that have a non-zero probability of occurring.

# Why is it Defined This Way? (Normalization)

We define  $P(A | B) = \frac{P(A \cap B)}{P(B)}$  because we want the new conditional evaluation to act as a **fully valid probability measure** over the restricted space  $B$ .

## 1. Restricting to the active region:

- Since  $B$  is guaranteed to have occurred, any outcome in  $A$  can only occur if it lies in the intersection  $A \cap B$ .

## 2. Establishing a new "Sure Event":

- If  $B$  is now our effective sample space, we must mathematically force  $P(B | B) = 1$ .
- Division by  $P(B)$  acts as a **normalizing scaling factor**:

$$P(B | B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

## Properties: Satisfying Kolmogorov's Axioms

For a fixed event  $B$  where  $P(B) > 0$ , the conditional mapping  $P(\cdot \mid B)$  satisfies all three of Kolmogorov's Axioms.

### Axiom 1: Non-Negativity

For any event  $A \subseteq \mathcal{S}$ :

$$P(A \mid B) \geq 0$$

**Proof:** Since  $P(A \cap B) \geq 0$  (Axiom 1) and  $P(B) > 0$ , their quotient must be non-negative.

### Axiom 2: Certainty (Normalization)

For the entire sample space  $\mathcal{S}$ :

$$P(\mathcal{S} \mid B) = 1$$

**Proof:** Since  $\mathcal{S} \cap B = B$ , we have:

$$P(\mathcal{S} \mid B) = \frac{P(\mathcal{S} \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1 \quad \blacksquare$$

# Properties: Countable Additivity

## Axiom 3: Countable Additivity

For any infinite sequence of mutually exclusive events  $A_1, A_2, \dots$ :

$$P\left(\bigcup_{i=1}^{\infty} A_i \mid B\right) = \sum_{i=1}^{\infty} P(A_i \mid B)$$

**Proof:** Using distributive laws, the events  $(A_i \cap B)$  are also mutually exclusive.

$$\begin{aligned} P\left(\bigcup_{i=1}^{\infty} A_i \mid B\right) &= \frac{P((\bigcup_{i=1}^{\infty} A_i) \cap B)}{P(B)} = \frac{P(\bigcup_{i=1}^{\infty} (A_i \cap B))}{P(B)} \\ &= \frac{\sum_{i=1}^{\infty} P(A_i \cap B)}{P(B)} \quad (\text{by Axiom 3 on disjoint elements}) \\ &= \sum_{i=1}^{\infty} \frac{P(A_i \cap B)}{P(B)} = \sum_{i=1}^{\infty} P(A_i \mid B) \quad \blacksquare \end{aligned}$$

# The Multiplication Rule

By rearranging the conditional probability definition, we obtain a powerful tool to find the joint probability of two overlapping events.

## Theorem: The Multiplication Rule

For any two events  $A$  and  $B$ :

$$P(A \cap B) = P(B) \cdot P(A \mid B)$$

Equivalently, if we condition on  $A$  instead:

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

This rule is highly practical for multi-stage random processes where the second stage directly depends on the outcome of the first.

## Exercise: Calculating Conditional Probabilities

### Problem Statement

A pair of fair, six-sided dice are rolled, yielding a sample space  $\mathcal{S}$  of 36 equally likely outcomes. Let:

- Event  $A$ : The sum of the two dice is 8.
- Event  $B$ : Both dice show even numbers.

Calculate:

- (a)  $P(A)$  and  $P(B)$
- (b) The conditional probabilities  $P(A \mid B)$  and  $P(B \mid A)$

# Extending the Multiplication Rule (The Chain Rule)

The multiplication rule generalizes naturally to three or more dependent events.

## Theorem: The General Chain Rule

For any sequence of events  $A_1, A_2, \dots, A_n$ :

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid \bigcap_{i=1}^{n-1} A_i)$$

## Example: Sequential Drawing Without Replacement

- Suppose we draw 3 cards sequentially from a 52-card deck without replacement.
- Let  $A_i$  be the event of drawing an Ace on the  $i$ -th draw:

$$P(A_1 \cap A_2 \cap A_3) = \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50}$$

## Exercise: Multi-stage Dependent Events

### Problem Statement

An urn contains 5 red balls and 4 blue balls. Three balls are drawn sequentially, one by one, **without replacement**.

Find the probability that the balls are drawn in the exact sequence:

First: Red  $\rightarrow$  Second: Blue  $\rightarrow$  Third: Red

*Hint: Let  $R_i$  and  $B_i$  denote drawing a Red or Blue ball on draw  $i$ . Apply the Chain Rule.*

# Independence of Events

Two events are independent if the occurrence of one provides absolutely no informational update regarding the likelihood of the other.

## Definition

**Statistical Independence** Two events  $A$  and  $B$  are said to be independent if and only if:

$$P(A \cap B) = P(A) \cdot P(B)$$

## Informational Intuition:

- Knowing that  $B$  happened does not change the probability of  $A$ :

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = P(A)$$

- Similarly, knowing  $A$  happened does not change the likelihood of  $B$ :

$$P(B | A) = P(B)$$

## Exercise:

### Problem Statement

Let  $A$  and  $B$  be two events such that:

$$P(A) = 0.4, \quad P(B) = 0.5, \quad P(A \cup B) = 0.7$$

Are the events  $A$  and  $B$  independent? Justify your answer using the definition of independence.

### Problem Statement

Suppose  $A$  and  $B$  are two mutually exclusive (disjoint) events, and  $P(A) > 0$  and  $P(B) > 0$ . Are  $A$  and  $B$  independent? Explain why or why not.

## Mutual Independence of Multiple Events

When dealing with three or more events, pairwise independence (checking events two at a time) is **not** sufficient to guarantee full independence.

### Definition

**Mutual Independence** A collection of events  $A_1, A_2, \dots, A_n$  is *mutually independent* if for **every** possible subset  $A_{i_1}, A_{i_2}, \dots, A_{i_k}$  of these events, the probability of their intersection equals the product of their individual probabilities:

$$P\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k P(A_{i_j})$$

# Mutual Independence of Multiple Events

For exactly three events  $(A, B, C)$ , this requires four conditions:

- **Pairwise:**  $P(A \cap B) = P(A)P(B)$
- **Pairwise:**  $P(A \cap C) = P(A)P(C)$
- **Pairwise:**  $P(B \cap C) = P(B)P(C)$
- **Mutual:**  $P(A \cap B \cap C) = P(A)P(B)P(C)$

# Reversing Conditional Probabilities

In many applications, we know the conditional probability of an effect given a cause, but we actually observe the effect and want to deduce the probability of the cause.

## The Problem of Inversion

- We easily know  $P(\text{Symptom} \mid \text{Disease})$ .
- But a patient walks in with the symptom; we must find  $P(\text{Disease} \mid \text{Symptom})$ .
- How do we mathematically reverse conditioning to find  $P(B \mid A)$  if we only have  $P(A \mid B)$ ?

## The Base Case: Two Exhaustive Scenarios

Let us start with the simplest possible scenario: an event  $B$  either happens, or it does not ( $B^c$ ). These two events are **mutually exclusive** ( $B \cap B^c = \emptyset$ ) and **exhaustive** ( $B \cup B^c = S$ ).

For any target event  $A$ , we can decompose it into these two scenarios:

$$A = (A \cap B) \cup (A \cap B^c)$$

Since these two components are disjoint, their probabilities add:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

Applying the multiplication rule gives the **Simple Law of Total Probability**:

$$P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c)$$

## Bayes' Theorem (Simple Form)

Using our definition of conditional probability, we can now write the probability of  $B$  given that we observe event  $A$ :

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

By replacing the numerator with the multiplication rule, and the denominator with our simple Law of Total Probability, we get:

### Theorem

*Bayes' Theorem (Two-Event Form) For any two events  $A$  and  $B$  where  $P(A) > 0$  and  $0 < P(B) < 1$ :*

$$P(B | A) = \frac{P(A | B)P(B)}{P(A | B)P(B) + P(A | B^c)P(B^c)}$$

## Scaling Up: General Partitions

What if there are more than just two underlying scenarios? We can scale this logic up to any arbitrary number of mutually exclusive possibilities.

**Definition :** A collection of events  $B_1, B_2, \dots, B_n$  is a **partition** of the sample space  $\mathcal{S}$  if:

- 1 They are pairwise mutually exclusive:

$$B_i \cap B_j = \emptyset \quad \text{for all } i \neq j$$

- 2 Their union completely exhausts the sample space:

$$\bigcup_{i=1}^n B_i = \mathcal{S}$$

- 3 Each partition slice has a positive probability:  $P(B_i) > 0$ .

# The General Law of Total Probability

When an event  $A$  can occur under several distinct, disjoint scenarios  $B_1, \dots, B_n$ , we find its total probability by summing across all partitions.

## Theorem

*The Law of Total Probability (General Form) Let  $B_1, B_2, \dots, B_n$  be a partition of  $S$ . For any event  $A$ :*

$$P(A) = \sum_{i=1}^n P(A \cap B_i)$$

*Applying the multiplication rule, this is equivalent to:*

$$P(A) = \sum_{i=1}^n P(A \mid B_i)P(B_i)$$

## General Form of Bayes' Theorem

Combining the conditional probability definition with the general Law of Total Probability allows us to resolve the posterior probability for any specific scenario  $B_k$ .

### Theorem

*Bayes' Theorem (General Form) Let  $B_1, B_2, \dots, B_n$  be a partition of the sample space  $\mathcal{S}$ . For any event  $A$  where  $P(A) > 0$ , the posterior probability for any specific slice  $B_k$  is:*

$$P(B_k | A) = \frac{P(A | B_k)P(B_k)}{\sum_{i=1}^n P(A | B_i)P(B_i)}$$

*The denominator is simply the total probability  $P(A)$  expanded across all  $n$  possible options.*

# Anatomy of the Bayesian Terms

Each term in Bayes' Theorem plays a distinct role in statistical inference and updating beliefs.

$$P(B | A) = \frac{P(A | B)P(B)}{P(A)}$$

- **Prior Probability**  $P(B)$ : Our initial assessment of the likelihood of event  $B$  before observing the new evidence.
- **Likelihood**  $P(A | B)$ : The probability that the evidence  $A$  would occur given that scenario  $B$  is true.
- **Posterior Probability**  $P(B | A)$ : The updated probability of  $B$  after taking the evidence  $A$  into account.
- **Evidence**  $P(A)$ : The overall probability of observing the evidence  $A$  across all possible scenarios.

## Example: The Monty Hall Problem

### The Setup

You are on a game show and presented with three closed doors. Behind one door is a new car; behind the other two are goats.

- You select a door, say **Door 1**.
- The host (Monty Hall), who knows what is behind every door, opens another door, say **Door 3**, revealing a goat.

### The Dilemma

Monty then turns to you and asks:

*"Do you want to stick with Door 1, or switch to Door 2?"*

## Solving Monty Hall: Conditioning on the Initial Guess

Let  $G$  be the event that your initial guess is correct (you picked the car).  
Let  $G^c$  be the event that your initial guess is incorrect (you picked a goat).  
Let  $W_S$  be the event that you win the car by switching doors.

### Defining the Conditional Probabilities

Since there is 1 car and 2 goats, we know initially that  $P(G) = 1/3$  and  $P(G^c) = 2/3$ .

- $P(W_S \mid G) = 0$  (If you initially picked the car, switching guarantees you get a goat)
- $P(W_S \mid G^c) = 1$  (If you initially picked a goat, the host *must* reveal the other goat, meaning the remaining closed door is guaranteed to be the car)

# Solving Monty Hall: Conditioning on the Initial Guess

## Calculating the Total Probability of Winning by Switching

We want to find the unconditional probability of winning by switching,  $P(W_S)$ . Using the Law of Total Probability, we partition the sample space by our initial guess:

$$P(W_S) = P(W_S \mid G)P(G) + P(W_S \mid G^c)P(G^c)$$

Substituting our known probabilities:

$$P(W_S) = (0) \left( \frac{1}{3} \right) + (1) \left( \frac{2}{3} \right) = \frac{2}{3}$$

**Conclusion:** This rigorously proves that switching doubles your probability of winning, without needing to track individual door assignments.

# Introduction to Pólya's Urn

## The Concept of "Contagion"

While independent events (like coin tosses) have no memory, many real-world processes are self-reinforcing—success breeds success, or one event increases the likelihood of identical future events.

## Definition (Pólya's Urn Setup)

Imagine an urn initially containing  $b$  black balls and  $w$  white balls.

- ① We draw a ball at random and note its color.
- ② We return the drawn ball to the urn.
- ③ **The Reinforcement:** We also add  $c$  *additional* balls of the same color and  $d$  *additional* balls of the opposite color into the urn.

*Note:* We will be working assuming  $d = 0$ .

# The Mathematical Mechanism

By adding balls of the drawn color, the sample space changes dynamically, breaking independence. We must use conditional probability to track the system. Let  $X_i$  be the outcome of the  $i$ -th draw. **The First Draw:**

$$P(X_1 = \text{Black}) = \frac{b}{b + w}$$

**The Second Draw (Conditioned on the First):** If the first ball was Black, the urn now has  $b + c$  black balls and a total of  $b + w + c$  balls.

$$P(X_2 = \text{Black} \mid X_1 = \text{Black}) = \frac{b + c}{b + w + c}$$

## Joint Probability via Chain Rule

$$\begin{aligned} P(X_1 = B \cap X_2 = B) &= P(X_1 = B) \cdot P(X_2 = B \mid X_1 = B) \\ &= \left( \frac{b}{b + w} \right) \left( \frac{b + c}{b + w + c} \right) \end{aligned}$$

## Real-World Modeling:

We can map this purely mathematical urn to real-world dependent phenomena, such as local weather patterns during a monsoon season.

### Weather events

- **Black Balls ( $b$ ):** Rainy days
- **White Balls ( $w$ ):** Sunny days
- **Reinforcement ( $c$ ):** Atmospheric persistence

### Disease Spread

- **Black Balls ( $b$ ):** Infected
- **White Balls ( $w$ ):** Susceptible
- **Reinforcement ( $c$ ):** Contagious spread

# The Recurrence Relation (Law of Total Probability)

## Setting up the Second Draw

Let  $B_i$  and  $W_i$  be the events of drawing Black or White on draw  $i$ . To find  $P(B_2)$ , we condition on the two possible outcomes of the first draw:

$$P(B_2) = P(B_2 \mid B_1)P(B_1) + P(B_2 \mid W_1)P(W_1)$$

## Evaluating the Probability

Substitute the known initial and conditional probabilities:

$$P(B_2) = \left( \frac{b+c}{b+w+c} \right) \left( \frac{b}{b+w} \right) + \left( \frac{b}{b+w+c} \right) \left( \frac{w}{b+w} \right)$$

Combine the numerators over the common denominator:

$$P(B_2) = \frac{b(b+c) + bw}{(b+w+c)(b+w)} = \frac{b(b+c+w)}{(b+w+c)(b+w)} = \frac{b}{b+w}$$

## Exercise: Medical Diagnostics (Base Rate Fallacy)

### Problem Statement

A rare medical condition affects exactly 1% of a population. A diagnostic test is developed to screen for this condition:

- **Sensitivity** (True Positive Rate): If a person has the condition, the test yields a positive result (+) with probability 0.99.
- **False Positive Rate**: If a person does not have the condition, the test still mistakenly yields a positive result (+) with probability 0.02.

If a randomly selected person from the population tests positive (+), what is the probability that they actually have the condition?

*Define your events clearly: Let  $D$  be the event of having the disease, and let  $T^+$  be the event of testing positive.*

## Exercise 2: Spam Notifications

### Problem Statement

A digital user receives notifications across three primary channels: Email ( $E$ ), SMS ( $S$ ), and Instant Messaging ( $M$ ):

- 50% of all notifications arrive via Email, 30% via SMS, and 20% via Instant Message.
- The likelihood of a notification containing a harmful link ( $L$ ) differs by channel:
  - For Emails, only 1% contain harmful links.
  - For SMS, 4% contain harmful links.
  - For Instant Messages, 10% contain harmful links.

If a randomly received notification is flagged as containing a harmful link ( $L$ ), what is the probability that it arrived via Instant Messaging ( $M$ )?